Tutorial-2, problem 2b

(b)

$$L = T - V$$

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$r = l(t) = \alpha t$$

$$T = \frac{1}{2} m (\dot{l}^2 + l^2 \dot{\theta}^2)$$

$$= \frac{1}{2} m (\alpha^2 + l^2 \dot{\theta}^2)$$

$$V = -mgl \cos\theta$$

$$L = \frac{1}{2} m (\alpha^2 + l^2 \dot{\theta}^2) + mgl \cos$$

ENERGY : $E = \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} - L = \dot{\theta} [m l^2 \dot{\theta}] - L$

$$\Rightarrow E = \frac{1}{2} m l^2 \dot{\theta}^2 - \frac{1}{2} m \alpha^2 - mgl \cos\theta .$$

-ve sign

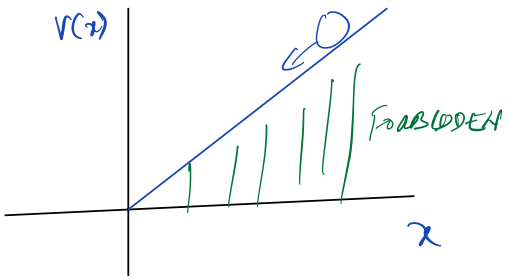
$$\Rightarrow E \neq T + V$$

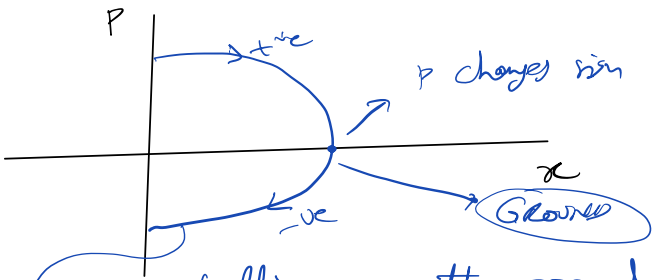
$\Rightarrow$ ENERGY IS NOT CONSERVED

Practice Problems-1, problem 3 b

$V(x) = Kx$

$E = \frac{1}{2} m v^2 - Kx$





Ball falling on the ground

Bouncing back